

Darshan Raja

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Education

The University of North Carolina at Charlotte , Charlotte , NC

January 2024 – December 2025

Masters in Information Technology, Concentration in Data Science | **GPA : 4.0**

Relevant Coursework: Data Mining | Mobile App Development | Information Privacy | Human Centered Design | AI in Healthcare | Database Systems | Applied Database

Kumaraguru College of Technology – Coimbatore, India

August 2019- April 2023

Bachelors in Information Science and Engineering | **GPA: 3.52**

Relevant Coursework: Java | DBMS | C++ | Computer Systems and Concepts | Algorithms & Data Structures | Web Development | Python Programming | Artificial Intelligence | Machine Learning

Professional Experience

SOFTWARE ASSOCIATE

February 2023– November 2023

INDIUM SOFTWARE PVT.LTD, CHENNAI

- ❑ Completed "Educational Consultancy Application" - Help students to reduce the hassle of finding a university will fully automated suggestions based on the users.
- ❑ Built automation scripts to integrate ECA with JIRA, emails, and other integrations software while working on lifecycle management.
- ❑ Got hands-on experience with load-balancers and applied automation to the same.
- ❑ Learned and applied numerous Machine Learning Technique and algorithms on developing a product.

Certification

- ❑ Flutter with Dart - <https://www.udemy.com/certificate/UC-fe614479-78db-4dab-bef4-44a9554e7a20/>
- ❑ Fundamentals of Visualization - <https://coursera.org/share/3b5a8b28c2ca63775d7c3a571fa3536d>
- ❑ Migrating to AWS Cloud - <https://coursera.org/share/c4eb54fc9bf5a47d87758297713e97ba>
- ❑ Agile with Atlassian JIRA - <https://coursera.org/share/161b6e9d68f337598cbe01995abe6f79>
- ❑ Machine Learning with Python - <https://coursera.org/share/b60665ff43d46992241bd50e84acb445>
- ❑ Software Security - <https://coursera.org/share/e4f5b85115ccb9687521cfb3d0b61791>

Technical Skills

Programming Languages: C, C++ , C# ,Java, Python, Swift, SQL

Web Development: HTML, CSS, JavaScript, Bootstrap

Database Management: MySQL , MongoDB, FireBase

Cloud: Aws

Other Tools/Frameworks: Spring Boot, Anaconda, IntelliJ, Pycharm, Junit, Colab

Project Tracking Tools: Jira

Academic / Personal Projects

LENGTH OF STAY PREDICTION IN ICU

August2024– December 2024

- ❑ Objective: Built predictive models to estimate ICU stay durations using MIMIC-IV datasets, focusing on patients expected to stay ≥ 7 days. Employed interpretability techniques to support clinical decisions and improve resource planning.
- ❑ Tools and Techniques: Python, Pandas, XGBoost, SHAP, LIME, PDP, Matplotlib, Seaborn, Sklearn, MIMIC-IV database

HEART RATE AND HEART DISEASE PREDICTION

August2024– December 2024

- ❑ Objective: Developed machine learning models to predict abnormal heart rates and diagnose heart disease using clinical datasets. Emphasized early detection to support preventive healthcare and reduce cardiac risk. Integrated domain-specific metrics to ensure accurate classification.
- ❑ Tools and Techniques: Python, Supervised Learning, Logistic Regression, Random Forest, Scikit-learn, Matplotlib, SHAP, LIME

DETECTION OF DRIVER ETIQUETTE FOR SAFE TRAVEL

March 2022– April 2023

- ❑ Objective: Mobile application system aims to address safety concerns by providing real-time monitoring of driver behavior and promptly notifying management of any abnormalities. It utilizes voice analysis and facial detection to detect driver drowsiness and instances of canceled rides, thereby enhancing passenger safety in the transportation industry.
- ❑ Tools and Technique: NLP, Python, Firebase Database, ML, CNN, Open-CV, Flutter Leads Scoring Case Study

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- <https://ieeexplore.ieee.org/document/10199231> - Publication

DIABETICS DETECTION

June 2022– September 2022

- Objective: The objective of utilizing Logistic Regression for diabetes prediction is to leverage a specialized Machine Learning algorithm that excels in classifying problems. This predictive analysis technique operates by utilizing probability to effectively determine the likelihood of an individual having diabetes or not.
- Tools and Technique: Python, ML, Supervised learning technique, HTML CSS, JS

TRAFFIC SIGNALS IMAGE PROCESSING

December 2021– March 2023

- Objective: Monitoring traffic cameras and issuing fines to violators is to enhance road safety, deter traffic violations, enforce traffic laws, promote fairness, generate revenue for road improvements, and provide data for traffic analysis.
- Tools and Technique: AI, CNN, Deep Learning, ANPR (Automatic Number Plate Recognition).

AUTOPARTS MANAGEMENT SYSTEMS

July 2021– November 2021

- Objective: The major goal is to develop an automated system that will enable the business owner to do all tasks with the push of a button and minimize the use of paper. Make a graphical user interface with a user-friendly interface and provide many login possibilities.
- Tools and Technique: Python, Tkinter - GUI, MongoDB Atlas

Hackathons

- Came up with a solution for a hackathon task that included creating a Blockchain-based graphic password authentication system. - March 2022
- Created and developed a system utilising NodeJS and MongoDB to coordinate the deployment and dispatch of trucks while monitoring their progress in real-time. - January 2023

Co- Curricular & Extra Curricular Activities

- HEAD OF VISUAL DESIGN - Visual Design team for a private organization as a part time member to build a creative design for digital marketing purposes.
- VP OF BIKER CLUB - Organized an event where participants rode their bikes for 30 km while displaying a poster saying "Save Trees" to raise awareness about the importance of environmental conservation, specifically focusing on the significance of preserving trees.
- CONNEXTIONS - Organized an event that brought together various visuals depicting coding methods, data structures, data types, and object-oriented programming, fostering connections and discussions among participants
- E-TRESSURE HUNT - Organized an event centered around computer puzzles, coding challenges, and advanced problem-solving techniques, with a focus on identifying the methods employed and strategies utilized for complex task resolution.
- SPORTS MINISTER – As Sports Minister, having successfully organized various sports events, including football, volleyball, and athletics. Additionally, effectively managed the entire team system during these events, ensuring smooth operation and coordination.